Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	anshrathore0202-cyber
Project Name	F1 Racing Intelligence Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution for the F1 Racing Intelligence Analysis project.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	**Data Access & Integration**	Ingest historical Formula 1 racing statistics, lap-by-lap telemetry, pit stop timings, driver standings, and constructor results from raw CSV/Ergast datasets into a centralized local workspace repository (`C:\Users\HP\Desktop\F1-Analysis`).	**High**
2	**Data Processing**	Perform automated data cleaning, normalization, null-value imputation, and schema transformation using Python ETL scripts to link driver IDs with circuit metadata and constructor scores seamlessly.	**High**
3	**Business Analysis**	Calculate key performance indicators (KPIs) such as average stint lap pace, qualifying-to-race position delta, pit stop efficiency, and multi-season constructor dominance scores.	**High**
4	**Business Insights & Analytical Outputs**	Provide interactive Tableau visualization dashboards showing season standings trends, head-to-head driver telemetry overlays, track performance maps, and strategy comparison summary cards.	**High**
5	**User Interaction & Analysis**	Enable users to dynamic filter and drill down statistics by Grand Prix season, circuit location, constructor team, specific driver, and wet/dry weather conditions.	Medium
6	**Data Update & Reporting**	Provide scheduled automated refresh routines to ingest new race weekend results and export high-resolution analytical PDF/PNG reports for presentation.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Tableau visual dashboard must feature an intuitive layout requiring zero training, with clear labels and interactive filters accessible within 2 clicks.
NFR-2	**Security**	The local dataset and code repository must be version-controlled securely on GitHub (`anshrathore0202-cyber/F1-Racing-Intelligence-Analysis`) with sanitized configuration paths.
NFR-3	**Reliability**	Data processing pipelines must handle dirty, missing, or malformed F1 telemetry inputs gracefully without halting system execution or corrupting stored metrics.
NFR-4	**Performance**	Interactive visual queries and dashboard filter updates must render in under 2 seconds for multi-season statistical queries containing over 100,000 telemetry rows.
NFR-5	**Availability**	The local analysis workspace and Tableau dashboard files must maintain 99.9% local operational uptime for offline data processing and report generation.
NFR-6	**Scalability**	The system architecture must support seamless expansion to ingest future Grand Prix telemetry data and additional motorsport series datasets without requiring structural redesign.